

BEHAVIORAL TREATMENT OF DEPRESSION IN A PREPUBERTAL CHILD

CYNTHIA FRAME

Department of Psychology, University of Georgia

JOHNNY L. MATSON

Department of Learning and Development, Northern Illinois University

WILLIAM A. SONIS

National Institute of Child Health and Human Development

and

M. JEROME FIALKOV and ALAN E. KAZDIN

Western Psychiatric Institute and Clinic, University of Pittsburgh School of Medicine

Summary—The present study evaluated behavioral treatment of symptoms of depression in a 10 yr-old boy. Diagnosis of the child's depression was made on the basis of DSM-III criteria. Information was obtained from separate interviews with the child and mother and from multiple assessment instruments. Ratings from several sources (mother, psychiatrist, psychologist and staff) confirmed the diagnosis. Four behaviors that characterized the child's depression were selected for intervention and included inappropriate body position, lack of eye contact, poor speech and bland affect. Treatment, evaluated in a multiple-baseline design across symptoms, consisted of the combination of instructions, modeling, role-playing and feedback. Results indicated that behaviors characteristic of childhood depression could be reliably identified and effectively treated by behavioral techniques. Treatment effects were maintained at 12-week follow-up assessment.

Childhood depression has been an active source of controversy and empirical investigation in recent years (Cantwell and Carlson, 1979; Puig-Antich, 1980; Schulterbrandt and Raskin, 1977). Evidence has emerged that depression in children can be reliably diagnosed using Research Diagnostic Criteria (RDC) or DSM III (e.g. Carlson and Cantwell, 1980; Puig-Antich *et al.*, 1979; Kashani, Barbero and Bolander, 1981). In addition, several techniques have been developed to diagnose depression and to measure its severity (Kovacs, 1981; Kazdin, 1981; Petti, 1978).

A growing body of literature for adults has suggested that behavior therapy may be useful to treat depressive disorders, particularly non-psychotic, unipolar depression (Rehm, 1981).

However, treatment of depressed children has yet to be evaluated systematically with psychosocial or behavioral techniques. The present investigation reports the case of a 10 yr-old child diagnosed as meeting criteria for a major depressive disorder. Behavioral treatment was conducted on symptoms of depression the child manifested in daily living.

METHOD

Subject

Dale was a 10 yr-old male inpatient of borderline mental retardation (full scale WISC-R IQ = 79) on a children's psychiatric intensive care unit. The inpatient facility houses 18 children who are admitted on a short-term basis for their acute psychopathology. Dale was admitted because of his recent history of suicidal thoughts and gestures, violent temper outbursts, and poor school performance. He was not on medication at any time during his hospitalization.

Requests for reprints should be addressed to: Alan E. Kazdin, Department of Psychiatry, Western Psychiatric Institute & Clinic, 3811 O'Hara Street, University of Pittsburgh School of Medicine, Pittsburgh, PA 15261, U.S.A.

Assessment

The subject was assessed with several different measures of depression. First, a psychiatric interview revealed that Dale met DSM III criteria for a major depressive episode. Second, upon admission, Dale's mother completed several measures including the Children's Depression Inventory (CDI) (parent-rated) (Kovacs and Beck, 1978), the Child Behavior Problem Checklist (CBP) (Achenbach, 1978), and the Bellevue Index of Depression (BID) (Petti, 1978). On each measure, Dale met criteria considered significant for depression. Specifically, Dale received a score of 27 on the CDI rated by his mother; scored one and one-half standard deviations above the mean for depression on the CBP; and met the Weinberg criteria (Brumback, Dietz-Schmidt and Weinberg, 1977) for depression on the BID.

The diagnosis was corroborated further by a rater who was naive to Dale's diagnosis and admitting problems. The rater viewed a video-taped interview of Dale and completed the CDI and Depression Adjective Checklist (DACL) (Lubin, 1967) for the child. The rater's judgments placed Dale in the clinically depressed range on the CDI with a score of 21, and at the 71st percentile for depression according to the DACL. Finally, reports by hospital staff revealed that Dale was frequently uncommunicative in interpersonal situations, with poor eye contact, inaudible speech or one-word answers, bland facial expression, and body positions indicative of social withdrawal, such as turning his head away from a speaker or placing his hands over his face. These last four areas were targeted for behavioral intervention.

Assessment was conducted daily while the subject responded to different situations in simulated role-play. The sessions were held on the inpatient unit in a treatment room with an observation booth separated by a one-way mirror through which the sessions were video-taped for later coding.

Target behaviors

The following behaviors were identified for assessment and treatment:

1. *Inappropriate body position.* The subject assuming any body position which indicated lack of interest or participation in the conversation during a response to a question. Examples included turning the head away, placing his hands over his face, bending forward or away from the interviewer, or standing with his back to the interviewer.
2. *Lack of eye contact.* The patient failing to look the interviewer in the eye at any time while delivering a verbal response.
3. *Poor speech quality.* This category was scored for occurrence if the subject: (a) uttered one or more words which were too soft or garbled to hear clearly, (b) spoke two or fewer words in response to a question or replied: "I don't know", or (c) responded to a question with a latency of 3 sec or longer.
4. *Bland affect.* Bland affect was defined as lack of emotional tone and inflection in the voice, and failure to employ facial, hand, or arm gestures, while speaking.

Each behavior was rated as occurrence or non-occurrence during the child's response to each of the 12 role-play situations presented one at a time, within a daily session. Thus, for each behavior, the child's score could range from 0 to

12 per day, where zero represented non-occurrence of the target behavior. Scenes involved typical everyday situations from home or hospital life, such as:

You are in the lunch line. A new girl is right behind you.

A staff member says, "Dale, tell the new girl how to earn dessert".

Assessment periods were identical for baseline, treatment, and follow-up, and occurred immediately after training during the treatment conditions.

Reliability of assessment. Ratings were made by a doctoral-level psychology intern and an undergraduate psychology student who were trained by a third person to a criterion of 80-90% agreement on all target behaviors. The student was "blind" to the experimental conditions and observed the video-taped sessions in random order.

Reliability was assessed by computing a Pearson product-moment correlation coefficient between the raters' session scores for each behavior. Eighty-six per cent of the sessions were rated by both observers, divided equally among baseline and treatment conditions. Agreement was high for each behavior: $r = 0.99$ for inappropriate body position, $r = 0.94$ for poor eye contact, $r = 0.83$ for poor speech, and $r = 0.95$ for bland affect.

Experimental conditions

The treatment was evaluated in a multiple-baseline design across each of three phases. The intervention consisted of a behavioral training package designed to alter the specific responses which characterized Dale's social interactions. The experimental design included three assessment periods; namely, baseline, treatment and follow-up.

Baseline. During baseline sessions, behavior was assessed in the manner described above. However, no attempt was made to intervene to alter performance. The durations of the baseline phase differed for each behavior to meet the requirements of a multiple-baseline design.

Treatment. Treatment was administered individually to the child by a psychology intern who served as the therapist. Twenty-minute sessions were held each weekday in the same room as described for assessment. Treatment consisted of a skills-training package involving instructions, modeling, role-play and performance feedback. Both assessment and novel role-play scenes were used for training. The child was told at the beginning of each session what constituted the inappropriate behavior to be trained that day, and a more appropriate response was modeled for him by the therapist. The child was then asked to perform a role-play scene. Appropriate responses were praised and followed by the presentation of a new scene. Inappropriate behaviors were immediately identified, and the child was required to repeat the role-play scene until his response was altered and acceptable, at which time the next scene was introduced. This procedure was repeated for the duration of the 20 min session.

After an 8-day baseline for all behaviors, the subject was trained simultaneously on the first two behaviors, body position and eye contact, for six sessions. This was followed by five sessions for the treatment of speech quality and ended with nine sessions teaching the child to express affect appropriately. The subject was given a small reward (candy or chewing gum) at the end of each daily session, contingent upon cooperation during the session rather than quality of response.

Follow-up. At the end of treatment the child was discharged from the hospital. Twelve weeks after treatment, the child returned to the hospital for reassessment. The conditions of assessment were identical to those described in the baseline phase.

RESULTS

The mean occurrence of each of the target behaviors across baseline, treatment and follow-up phases for the 12 role-play situations are presented in Table 1. Daily performance is illustrated in Fig. 1 which shows that treatment produced marked changes in behavior. The first two behaviors treated were inappropriate body position and poor eye contact. Although considerable variability was evident during baseline, these behaviors were high rate and either stable or increasing in the opposite direction of the desired changes. Both of the behaviors were rapidly modified in the appropriate direction with the introduction of treatment. Inappropriate body position and poor eye contact were consistently near zero at the conclusion of the treatment phase.

Table 1. Mean frequency of each of the four target behaviors by experimental condition

	Baseline	Treatment	Follow-up
Inappropriate body position	9.3	0.9	4.0
Poor eye contact	10.3	2.4	4.0
Poor speech quality	6.9	0.7	2.0
Bland affect	7.0	0.3	0.0

The third behavior treated was speech quality. During baseline, which extended well into the treatment phase for the first two behaviors treated, no appreciable change of the target behaviors was noted. However, when treatment began, rapid reductions to near zero responding were noted. Response rates were maintained at this level throughout the remainder of the treatment condition. The most variable behavior in baseline was bland affect, which was the final behavior treated. Despite this variability, the inappropriate behavior at its lowest rate during baseline was much higher than at any point following the inception of treatment.

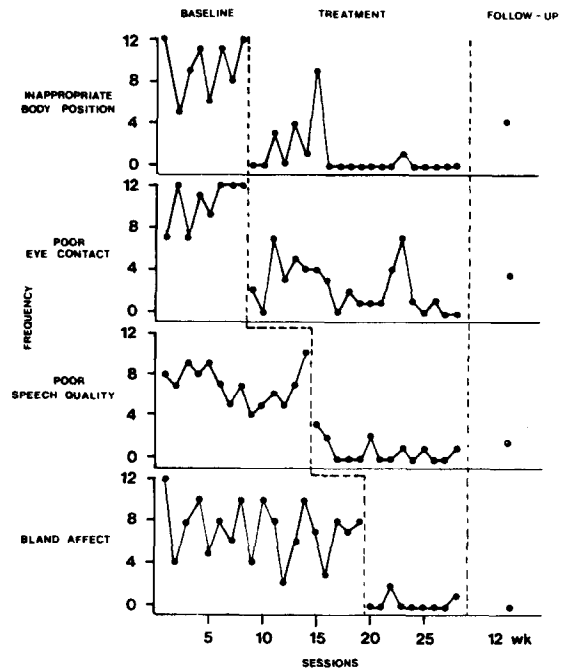


Fig. 1. Frequency of target behaviors for the 12 role-played scenes during baseline, treatment and follow-up.

In general the pattern of the data is relatively clear. Each behavior systematically improved when treatment was introduced, thus meeting the criteria of the multiple-baseline design. The effects achieved with treatment continued to be evident 12 weeks later when the child returned to the hospital for follow-up assessment. At follow-up some increases were evident, especially for inappropriate body positions, but all behaviors remained well below the baseline rates (see Table 1 and Fig. 1).

DISCUSSION

The present study suggests that behaviors characteristic of depression can be identified, reliably assessed, and treated successfully in children. The present use of multiple measures and the reliance upon several sources of information greatly enhanced confidence in the general diagnosis of depression given to the child. There was consensual validation that

depressive behaviors were present and problematic and that these behaviors formed a symptom cluster typical of a depressive disorder. This suggests that the diagnosis did not result from artifact of particular assessment devices or bias of a particular observer. The demonstration also is consistent with the growing body of evidence that depression in childhood is an identifiable psychiatric entity.

The treatment of selected depressive behaviors was quite successful in this study. Changes in undesirable behaviors were immediate, marked and durable. Demonstration of treatment effects in a multiple-baseline design ensured that improvement was not attributed to extraneous factors, since each target behavior failed to decrease until it was subjected to treatment. These results are consistent with those of Matson (1981), who successfully reduced the frequency of depressive behaviors in four mentally retarded adults. To the authors' knowledge, the present study is one of the first demonstrations of the efficacy of behavior therapy with a child for behaviors characteristic of depression. Therapeutic changes were achieved without the use of medication. Because of the undetermined efficacy and side effects of antidepressants in children, an effective behavioral approach for depressive behaviors would provide an especially valuable treatment alternative.

The present study represents an encouraging first step toward the investigation of behavioral treatment of depression in children. However, much additional research is needed. One concern is the inability of the present study to permit analysis of the effectiveness of the individual treatment components.

This issue may be minor in light of the demonstrated efficacy of the package, as well as findings by other investigators that the combined treatment package is generally superior to isolated techniques in the training of social behaviors (Bellack and Hersen, 1979). A second issue pertains to the generalization of therapeutic change. Although it is valuable to know that depressive behaviors can be controlled

within the treatment setting, evidence for generalization of appropriate behaviors to daily life situations would be highly desirable.

The most important limitation of the present study is the inability to demonstrate elimination of a depressive disorder. The target symptoms chosen for treatment were an important part of this child's depressive symptom picture, but were neither pathognomonic for, nor unique to, depression. There is no evidence that the other facets of the depressive disorder (vegetative signs, for example) were altered by the treatment. Although multiple symptoms of the child's depression were successfully treated, the extent to which treatment eliminated the disorder remains unclear.

REFERENCES

- Achenbach T. M. (1978) The Child Behavior Profile: I. Boys aged 6-11, *J. Consult. Clin. Psychol.* **46**, 478-488.
- Bellack A. S. and Hersen M. (Eds.) (1979) *Research and Practice in Social Skills Training*. Plenum Press, New York.
- Brumback R. A., Dietz-Schmidt S. G. and Weinberg W. A. (1977) Depression in children referred to an educational diagnostic center: Diagnosis and treatment and analysis of criteria and literature review, *Dis. Nerv. Syst.* **38**, 529-535.
- Cantwell D. P. and Carlson G. (1979) Problems and prospects in the study of childhood depression, *J. Nerv. Ment. Dis.* **167**, 522-529.
- Carlson G. A. and Cantwell D. P. (1979) A survey of depressive symptoms in a child and adolescent psychiatric population, *J. Am. Acad. Child Psychiat.* **18**, 587-599.
- Kashani J. H., Barbero G. J. and Bolander F. D. (1981) Depression in hospitalized pediatric patients, *J. Am. Acad. Child Psychiat.* **20**, 123-134.
- Kazdin A. E. (1981) Assessment techniques for childhood depression: A critical appraisal, *J. Am. Acad. Child Psychiat.* **20**, 358-375.
- Kovacs M. (1981) Rating scales to assess depression in school aged children, *Acta Paedopsychiatrica* **46**, 305-315.
- Kovacs M. and Beck A. T. (1978) Maladaptive cognitive structures in depression, *Am. J. Psychiat.* **135**, 525-533.
- Lubin B. (1967) *The Depression Adjective Checklist, Form A*. Educational and Industrial Testing Service, San Diego.
- Matson J. L. (1982) The treatment of behavioral characteristics of depression in the mentally retarded, *Behav. Ther.* **13**, 209-218.
- Petti T. A. (1978) Depression in hospitalized child psychiatric patients: Approaches to measuring depression, *J. Am. Acad. Child Psychiat.* **17**, 49-59.

- Puig-Antich J. (1980) Affective disorders in childhood: A review and perspective, *Psychiat. Clin. North Am.* 3, 403-424.
- Puig-Antich J., Chambers W., Halpern F., Hanlon C. and Sachar E. J. (1979) Cortisol hypersecretion in prepubertal depressive illness: A preliminary report. *Psychoneuroendocrinology* 4, 191-197.
- Rehm L. P. (Ed.) (1981) *Behavior Therapy for Depression: Present Status and Future Directions*. Academic Press, New York.
- Schulterbrandt J. G. and Raskin A. (Eds.) (1977) *Depression in Children: Diagnosis, Treatment and Conceptual Models*. Raven Press, New York.